PREDICTING EMPLOYEE TURNOVER: A DATA ANALYSIS OF HR EMPLOYEE CHURN

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RESEARCH QUESTION & HYPOTHESIS

Q: What factors drive voluntary employee turnover, and how can analytics inform retention strategies?

H: Satisfaction, salary, and tenure are the primary predictors — low-satisfaction and low-salary employees show the highest attrition rates.

DATASET

HR Employee Churn Dataset · 14,999 records · 10 variables

Variable	Type	Range
satisfaction_level	Float	0.09–1.00
last_evaluation	Float	0.36–1.00
number_project	Integer	2–7
avg_monthly_hours	Integer	96–310
time_spend_company	Integer	2–10 yrs
salary	Categorical	low/med/high
left (target)	Binary	0 = stayed, 1 = left

METHODS

- **Cleaning:** imputation, outlier detection, type validation
- **Stats:** mean, median, SD, quartiles, Pearson r
- **Testing:** t-test & chi-square for significance
- **Tools:** Python · Pandas · Matplotlib · SciPy

FIG 1 — TURNOVER DISTRIBUTION

3,571 of 14,999 employees (23.81%) voluntarily left — 1 in 4, well above the 10–15% industry benchmark. Low-salary employees and those with 3–5 years of tenure accounted for the highest share of departures.

FIG 2 — WORKLOAD & ATTRITION

Attrition is **bimodal**: departures cluster at **~140 hrs/month** (disengaged) and **~250+ hrs/month** (burned out). Employees who stayed showed a moderate distribution around 200 hrs/month — both extremes are retention risks.

14,999	23.81%	−0.388	4.5×
Employee Records	Overall Turnover Rate	Satisfaction–Turnover Correlation	Higher turnover: Low vs. High salary

INTRODUCTION

Replacing one employee costs **50–200% of their annual salary** in recruiting and lost productivity — making retention a direct business priority.

This study analyzes 14,999 HR records to uncover the primary drivers of voluntary departure using descriptive statistics, correlation analysis, and visualization.

RESULTS — VISUALIZATIONS

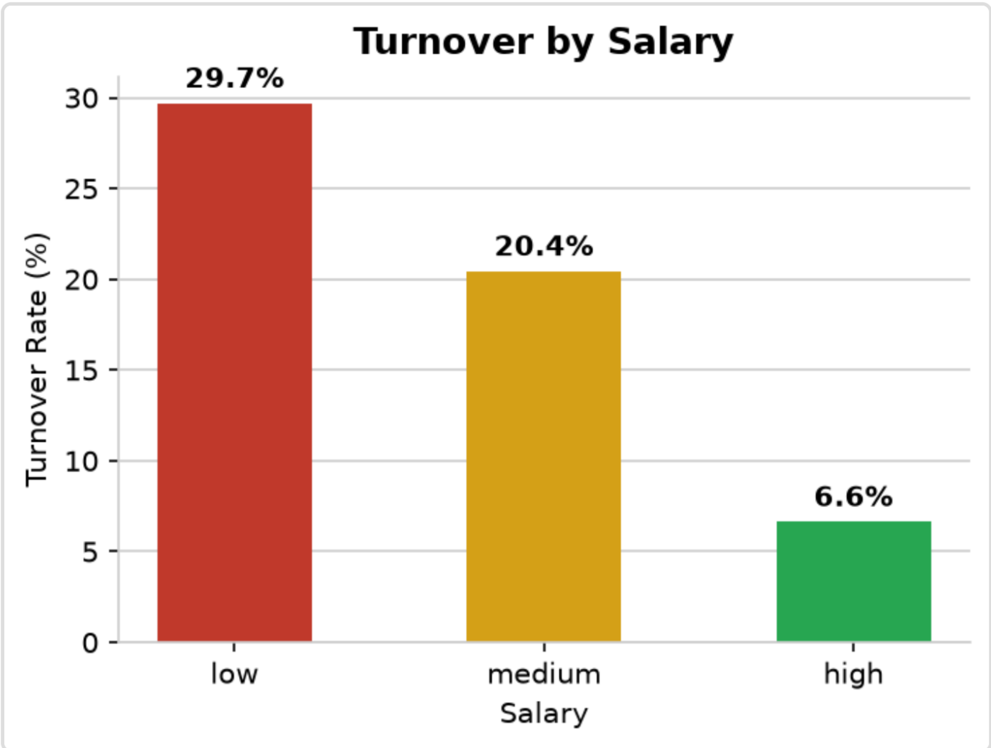

Fig 1. Turnover distribution — 3,571 (23.81%) left

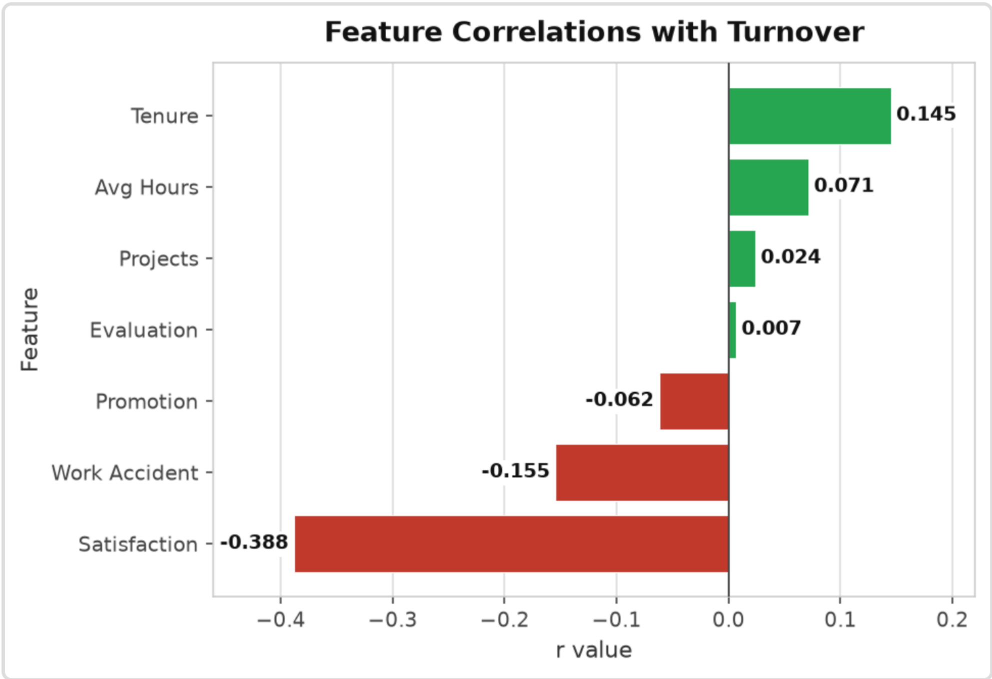

Fig 3. Avg satisfaction: 0.667 (stayed) vs. 0.440 (left)

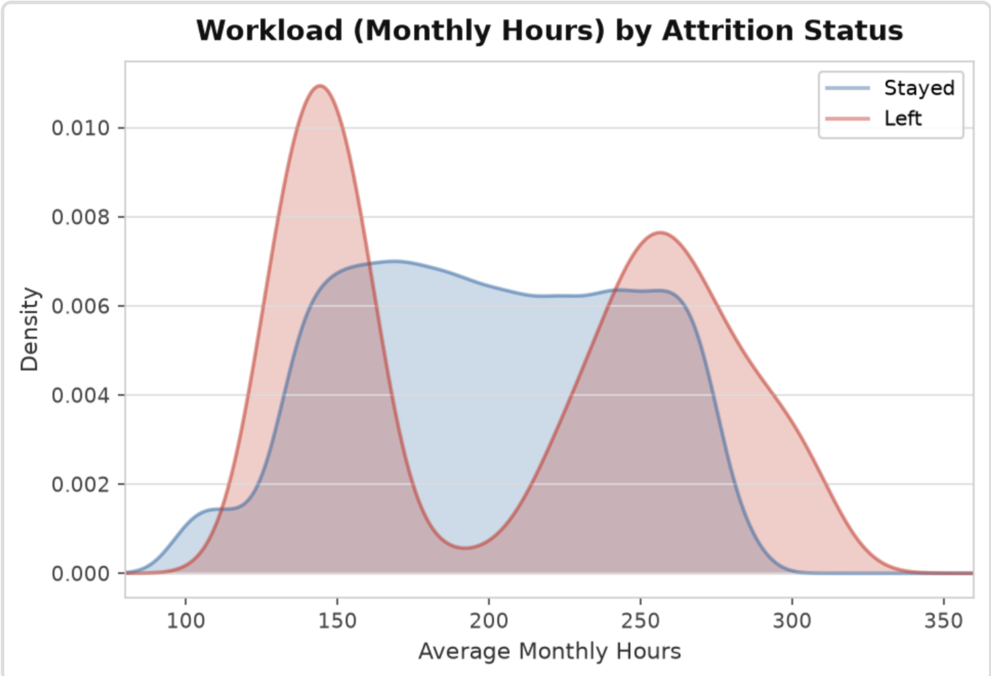

Fig 2. Workload distribution — employees who left cluster at ~140 hrs (disengaged) or ~250+ hrs (burned out)

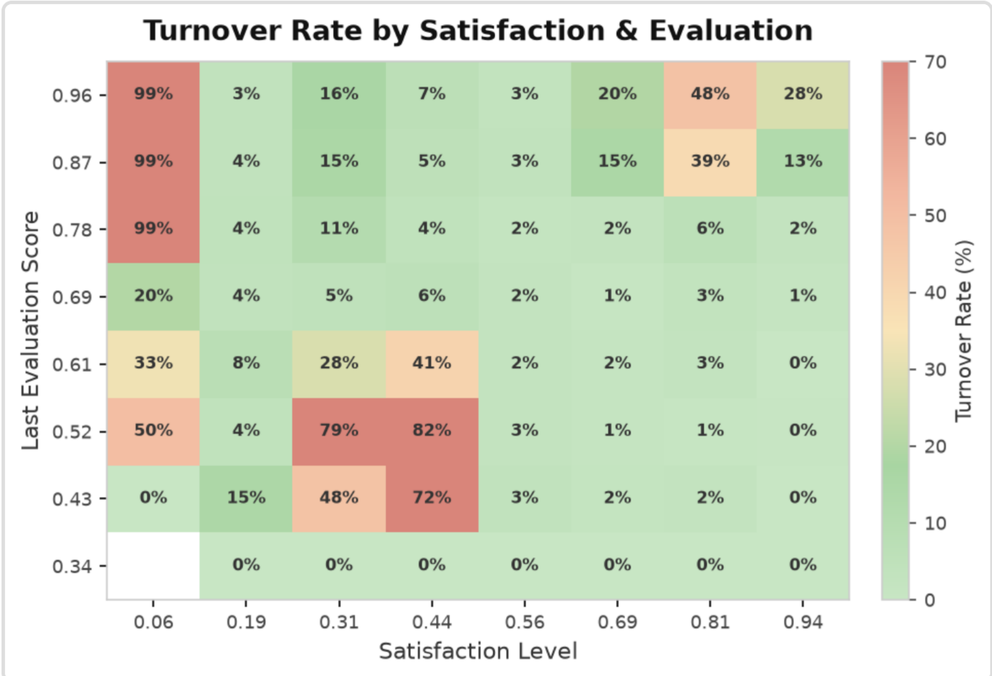

Fig 4. Turnover rate heatmap — high attrition concentrated in low-satisfaction cells regardless of evaluation score

Hypothesis: ✓Accepted — All three predictors significant at $p < 0.001$ — satisfaction ($t = -51.58$), salary ($\chi^2 = 381.23$), tenure ($r = 0.145$).

CONCLUSION

- **Low satisfaction** — dominant predictor ($r = -0.388$); invest in pulse surveys
- **Low compensation** — 4.5× higher attrition vs. high salary
- **Mid-career tenure (3–5 yrs)** — highest-risk window; mentorship recommended

KEY FINDINGS

Finding 1 — Turnover Rate: 23.81% left — 1 in 4 — well above the 10–15% industry benchmark.

Finding 2 — Satisfaction: Left avg **0.440** vs. stayed **0.667** ($r = -0.388$) — the strongest predictor.

Finding 3 — Salary: Low-salary turnover **29.7%** vs. 6.6% high-salary — a 4.5× gap.

Finding 4 — Tenure: 3–5 year tenure is the highest-risk attrition window ($r = 0.145$).

FIG 3 — SATISFACTION BY TURNOVER

Left employees averaged **0.440** vs. **0.667** for stayed ($r = -0.388$) — the **strongest predictor**. Scores below **0.5** signal elevated risk. Pulse surveys are the highest-return HR intervention.

FIG 4 — TURNOVER RATE HEATMAP

Low satisfaction (<0.4) drives 40–60%+ turnover regardless of evaluation score — even top performers leave when dissatisfied. **High satisfaction (>0.7)** drops turnover to single digits, confirming satisfaction as the dominant attrition axis.

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FUTURE RESEARCH

- Apply **ML models** (random forest, XGBoost) for predictive attrition classification
- Incorporate **department & role** for sector-level analysis
- Test on real-world datasets to validate generalizability

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